Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, product preferences, and subscription behavior from **3,900 transactions** to guide strategic business decisions.

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Dataset at a Glance

3,900

Purchases

Rows of transactional data

18

Features

Columns per record

37

Missing

Null values in Review Rating

Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Review Rating, Shipping Type

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Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean & Standardize

Imputed missing Review Ratings using category median; renamed columns to snake_case; dropped redundant `promo_code_used`.

03

Feature Engineering

Created `age_group` bins and `purchase_frequency_days` column.

04

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for SQL analysis.

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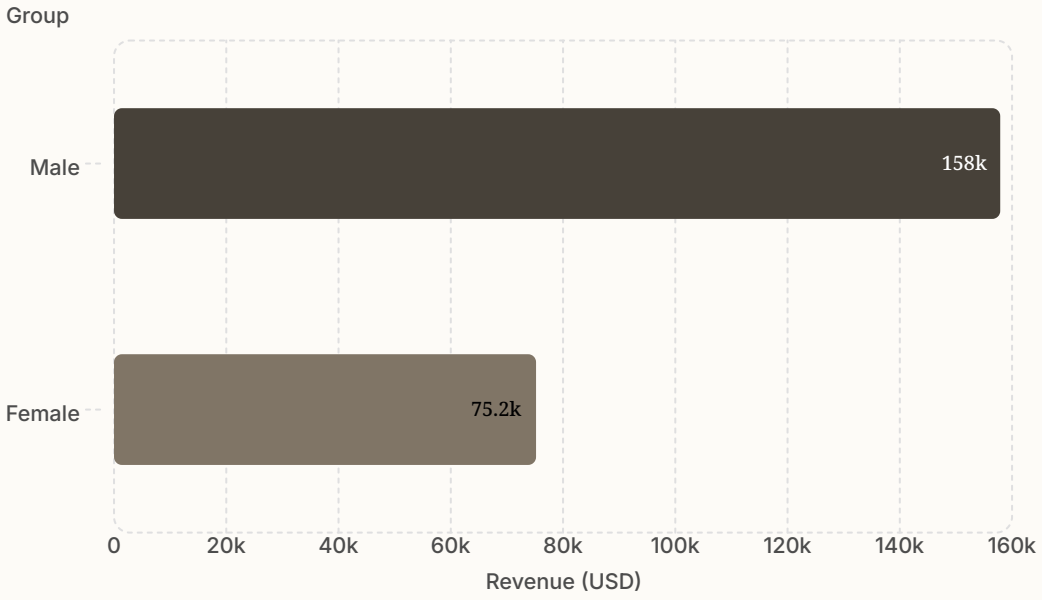
Revenue & Spending Insights



Shipping & Spend

Shipping Type	Avg. Purchase (USD)
Standard	$58.46
Express	$60.48

Express shipping users spend ~**$2 more** on average — a signal for upsell targeting.

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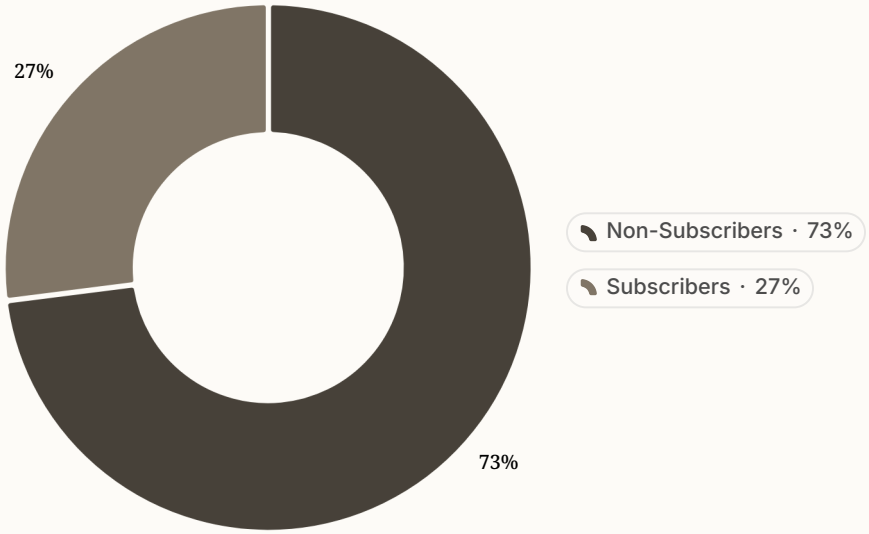
Subscriber vs. Non-Subscriber Behavior



Spend Comparison

Status	Customers	Avg. Spend	Total Revenue
Subscribers	1,053	$59.49	$62,645
Non-Subscribers	2,847	$59.87	$170,436

Average spend is nearly identical — subscription drives volume, not higher individual spend.

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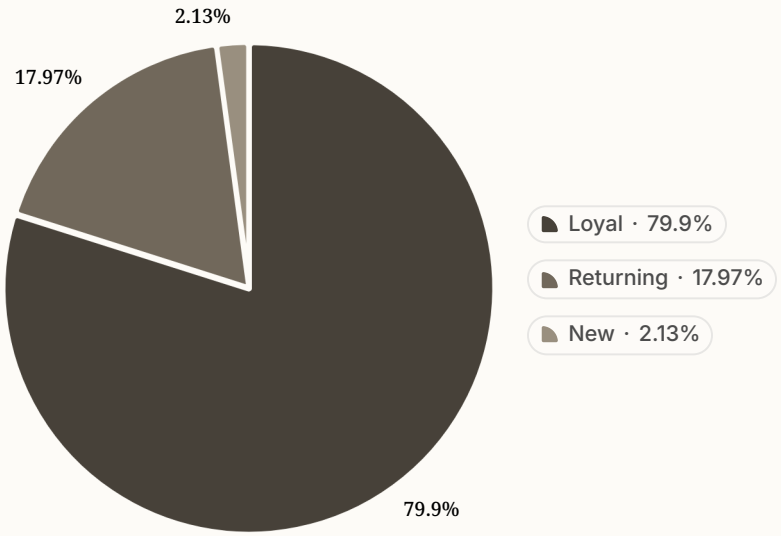
Customer Segmentation



Repeat Buyers & Subscriptions

Among customers with **more than 5 purchases**:

Subscription Status	Repeat Buyers
No	2,518
Yes	958

The vast majority of the base is already **Loyal** — focus on converting Returning customers.

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Product Performance

Top 5 Products by Rating

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

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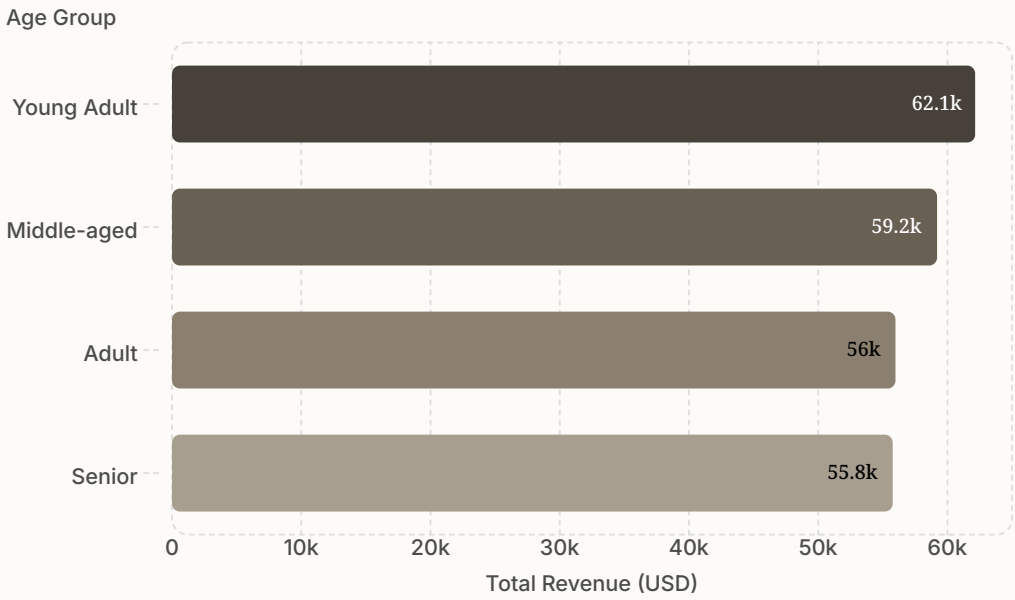
Revenue by Age Group



Top Products by Category

Category	#1 Product	#2 Product
Accessories	Jewelry (171)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)
Footwear	Sandals (160)	Shoes (150)
Outerwear	Jacket (163)	Coat (161)

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Power BI Dashboard

3.9K

Customers

$59.76

Avg. Purchase

3.75

Avg. Rating

Interactive filters by Subscription Status, Gender, Category, and Shipping Type. Visualizations include revenue and sales by category and age group, plus subscription breakdown.

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to grow the 27% subscriber base.

Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment.

Review Discount Policy

Balance sales boosts with margin control on high-discount products.

Targeted Marketing

Focus on Young Adults and Express-shipping users — highest revenue segments.

Product Positioning

Feature top-rated and best-selling products prominently in campaigns.

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